7.3 CNN Text Classification

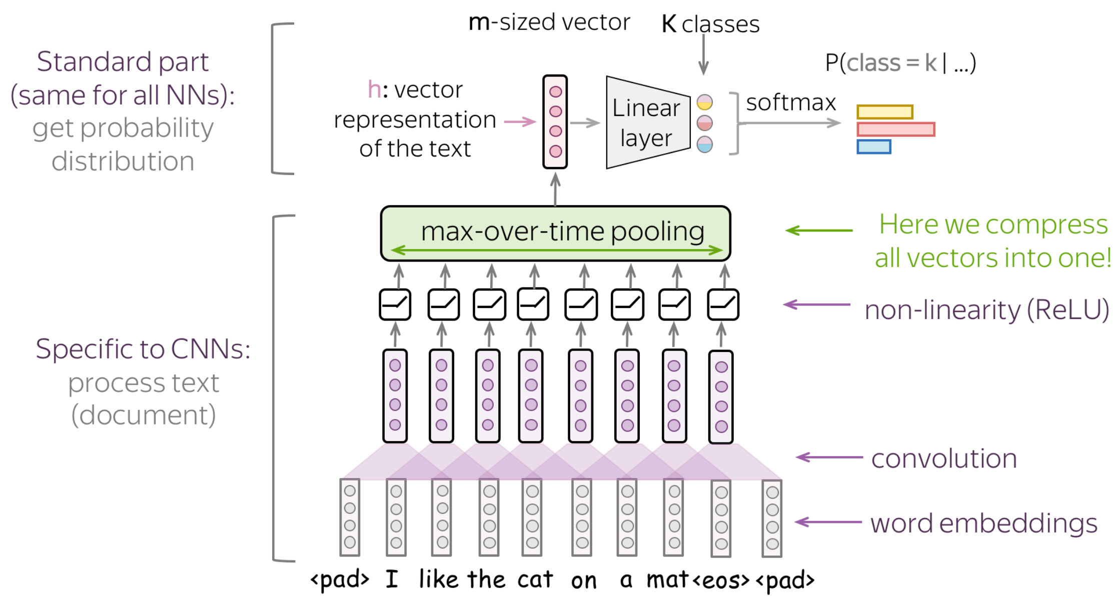
Several kernel sizes, then a classifier on the pooled features.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

1. From a sentence to one vector

Note **7.2** gave you 1-D filters over embeddings. Classification needs a **single** vector (or a small stack) that you can pass to a softmax.

The basic move: run the convolutions, then **global pool over time**. Each filter becomes one number (its max, or its mean, over every window). A sentence of any length becomes a vector of length `n_filters`. Then a dense layer plus softmax (or sigmoid) produces the label.



The middle can change—more layers, dropout, a residual—but **somewhere** you have to collapse time. If you skip that collapse you still have a sequence, which is a tagging setup, not a document label.

Sources: *Practical Natural Language Processing*, *Natural Language Processing in Action*, and Voita's notes.

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2. Why global max is the default

Global **max** asks: did this detector fire **anywhere**? For sentiment, topic, and spam that is usually the right question. *terrible* in position 2 or position 40 should both count.

Global **mean** asks how often / how strongly it fired. Use it when the label depends on the whole document (average tone of a long review), not on one smoking-gun phrase.

Either way you get a fixed-width representation. That is the CNN analogue of "average the word vectors," except the features are **learned phrases**, not single tokens.

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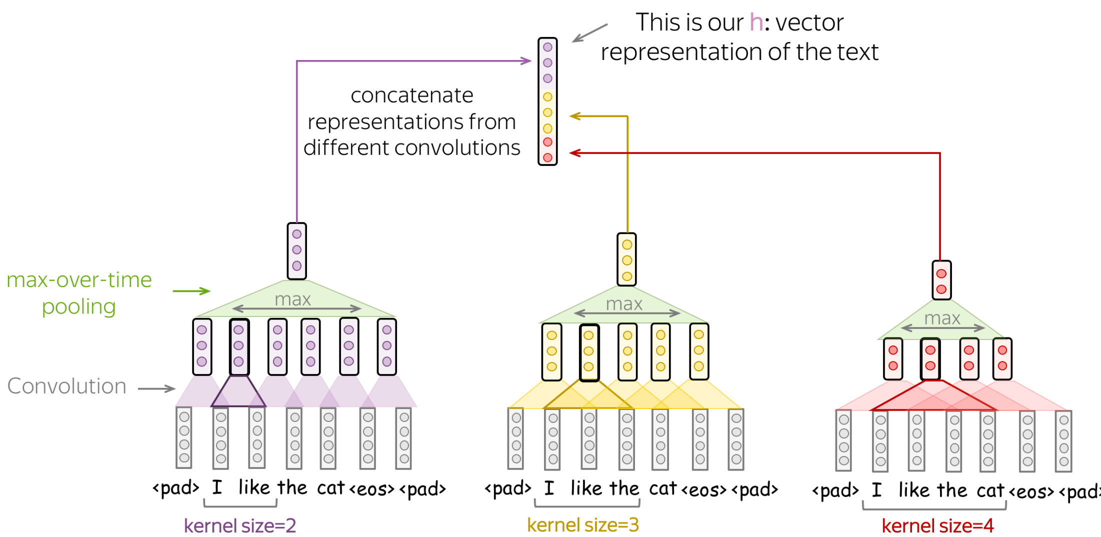
3. Several kernel sizes at once

A kernel of 2 sees pairs. A kernel of 3 sees short phrases. A kernel of 5 sees a longer idiom. You do not have to pick one.

Train **parallel** conv banks: 100 filters of size 2, 100 of size 3, 100 of size 5 (Kim 2014 is the paper everyone cites). Concatenate the three global-pooled vectors. Classify.



That is the architecture you should be able to draw. It is also the one you will see in Keras as three `Conv1D` branches into `Concatenate`.

Piece	Typical choice
Embeddings	100–300-D, pretrained or learned
Kernel sizes	2, 3, 4 and/or 5
Filters per size	64–128 on small data; more if you have it
Pool	global max
Regularization	dropout on the concatenated vector
Loss	binary or categorical cross-entropy

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4. How this sits next to Week 6

Week 6: TF–IDF + linear SVM. Fast, strong, no GPU.

This note: embeddings + multi-kernel CNN. You pay for training time. You gain detectors that share strength across similar phrases (*not good / not great*) and a representation whose width does not grow with $|V|$.

If the CNN does not beat the SVM by enough to matter, ship the SVM. The pipeline in note **6.1** did not change: vectorize (here: embed + conv), train, evaluate with F1 when the classes are skewed.

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5. Failure modes

- **Too few examples.** A CNN has more knobs than `LinearSVC`. On 500 labeled tweets the SVM usually wins.
- **Kernel too wide.** Size 15 on tweets is almost the whole sentence, once. You wanted local features.
- **No padding / tiny sentences.** Edge tokens never sit in the center of a valid window (note **7.1**).
- **Ignoring pretrained embeddings.** Random embeddings need more data. Start from Word2Vec or GloVe unless the domain is truly alien.

Week 8 replaces the fixed window with a **loop** that can carry information across the whole sentence.

Kim-style checklist. (1) Embed. (2) Three `Conv1D` branches. (3) Global max on each. (4) Concatenate. (5) Dropout. (6) Dense + softmax. If you cannot point to those six pieces in a notebook, you do not yet have the lecture model—you have a pile of layers.

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6. Evaluation stays Week 6

Accuracy if the labels are balanced. F1 if they are not. A CNN that is 2 points of accuracy better than an SVM and 10 points of minority-class F1 worse is not an upgrade. Report the same `classification_report` you used for k-NN.

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7. Practice

1. A review has 40 tokens and you use 128 filters of size 3 with global max. What is the width of the vector that enters the dense layer?
2. Why concatenate kernels of size 2, 3, and 5 instead of one kernel of size 5?
3. You have 400 labeled emails and a good TF–IDF SVM. Should you replace it with this CNN first, or report the SVM as the baseline? Why?